Pranshu Gupta

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Professional Experience

- **Microsoft SuperIntelligence** */ Member of Technical Staff / Redmond, WA* *Jul 2026 – Present*
  - Optimized **base model inference** in Multi-LoRA serving by bypassing adapters for base-only batches: **6.7% higher throughput** and **3.6% lower inter-token latency**, with no added synchronization overhead.
  - Benchmarked Multi-LoRA inference, profiled GPUs with PyTorch/Nsight, and tuned the CSGMV LoRA kernel.
  - Fixed LoRA adapter memory partitioning for attention data parallelism, enabling inference with heterogeneous layer shapes.
  - Extended checkpoint conversion for shared experts, mixed precision, heterogeneous attention layouts and vocabularies.
  - Built a LoRA fine-tuning pipeline combining supervised learning and reinforcement learning for shopping tasks, improving exact product-match accuracy from **28.3% to 37.2%**.
  - Standardized inference incident triage with a shared playbook and scoped telemetry queries for diagnosing inference issues.
- **Microsoft AI** */ Member of Technical Staff / Redmond, WA* *Aug 2025 – Jul 2026*
  - Reduced inference costs by eliminating non-essential classifiers (**20% lower GPU footprint**) and resolving token overhead in the Chat Completions API (**25% fewer input tokens** and **30ms lower time to first token**).
  - Implemented constrained decoding for tool invocation in the Completions API for GPT-5.1 instant and reasoning models.
  - Defined safety classification API contracts for biosafety risks and reasoning leakage, adopted by downstream teams.
  - Designed and shipped **incremental reasoning summarization**, progressively surfacing summaries in the UI.
  - Established M365 Copilot's shared LLM **traffic-classification taxonomy** (product, client app, inference intent, license tier), aligning teams and driving adoption through Grafana dashboards for capacity planning and cost attribution.
- **Microsoft Azure AI** */ Senior Software Engineer / Redmond, WA* *Jan 2025 – Aug 2025*
  - Designed and built the inference pipeline for an intelligent **model router**: injecting per-request model selection based on cost/complexity signals before each LLM call, reducing per-inference cost by **55%**.
  - Built a distributed rate limiter with OpenAI Tiktoken and output-token reconciliation, replacing a length-based estimator; actual inference-engine token usage averaged **219% above its estimates**.
- **Microsoft Azure Core** */ Senior Software Engineer / Redmond, WA* *Mar 2024 – Jan 2025*
  - Served as **technical area owner** for resource lifecycle management in Azure Resource Manager.
  - Independently designed and shipped **async operation URI signing**; ran office hours with resource provider teams to guide API surface changes, adopted by **242 service teams** with **220+ enforcing it**.
  - Designed and shipped a token-bucket-based distributed rate limiter for background jobs with retry-storm protection, preventing noisy-neighbor throttling in Azure Control Plane background-job infrastructure.
  - Mentored a new-grad engineer through weekly 1:1s; they now independently own a critical component in the control plane.
- **Microsoft Azure Core** */ Software Engineer II / Redmond, WA* *Jun 2020 – Mar 2024*
  - Designed and shipped **async operation callbacks** to replace polling, cutting control plane polling traffic by **94%**.
  - Led **Azure region teardown** design and implementation end-to-end: halted new resource creation, resolved subscription/resource group hierarchies, and migrated live resources to healthy regions with zero data loss.
  - Built dependency-aware deletion for Azure resource groups and bulk-deletion APIs, using staged retries and opt-in force deletion to reduce dependency-related deletion errors by **92%**.
  - Improved termination and retry policies for Azure resource consistency jobs with exponential backoff, random jitter, and resource-existence checks, reducing timeouts by **56%**, job duration by **72%**, and execution count by **70%**.
- **Microsoft Core Services** */ Software Engineer / Hyderabad, India* *Jun 2017 – Aug 2019*
  - Modernized a data enrichment platform to Azure Function Apps (**$4K/month** savings); integrated Cognitive Services for data standardization, and built a bulk-processing pipeline with **40–50x throughput** improvement.

Technical Skills

- **AI/ML & Inference**: Distributed LLM inference/serving, PyTorch, LoRA, GPU profiling (Nsight), checkpoint conversion
- **Languages & Frameworks**: Python, C#, C++, KQL, JavaScript, .NET
- **Systems & Cloud**: Kubernetes, Redis, Grafana; Azure (Cosmos DB, ARM Templates, Service Bus, DevOps)

Publications & Writing

- *Runtime Optimizations for LLMs* (2026). Production notes on inference optimizations for latency/cost-efficient LLM serving.
- *The Observer Effect on Social Media* — ACM CHI (2024). Causal inference and time series study on longitudinal Facebook data from 300+ participants to quantify how monitoring awareness alters online behavior.
- *Mental Health Impact of Active-Shooter Drills* — Humanities and Social Sciences Communications (2021). Applied ML classifiers and interrupted time series analysis to 54M social media posts across 114 schools to measure psychological impact.

Education

- **Georgia Institute of Technology** */ M.S. Computer Science (Machine Learning) / Atlanta, GA* *May 2020*
- **Indian Institute of Technology, Kanpur** */ B.Tech, C.S. and Eng. / Kanpur, India* *May 2017*